Advanced LLM Capabilities through RAG and Tools

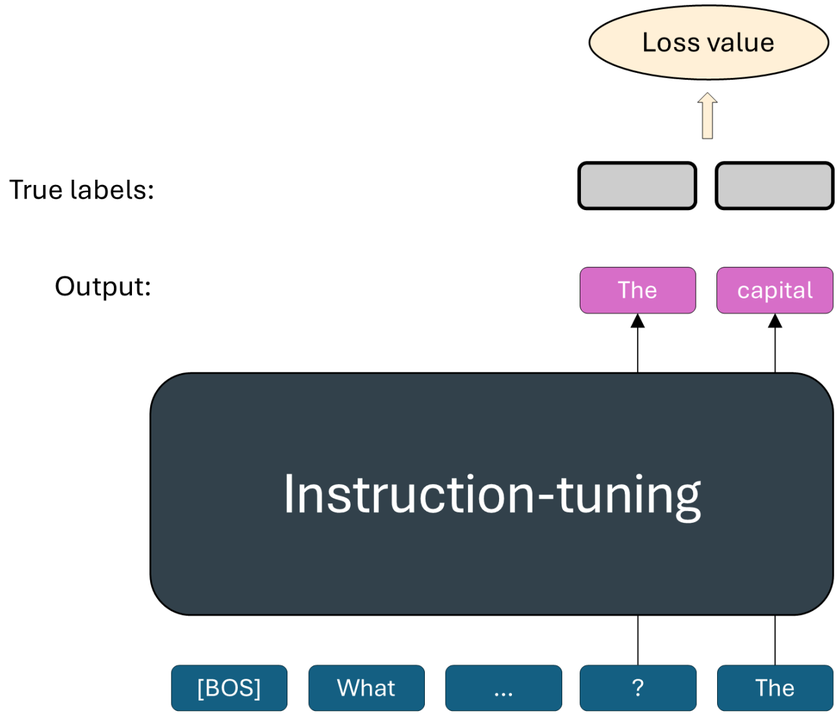
Anna Vettoruzzo

Overview

- Agents
- LLM agents
- Retrieval Augmented Generation
- Tool usage
- MCP protocol

LLM recap

- Produces output one token at the time based on the input prompt.
- Functions through next-token prediction.
- Requires a massive volume of data for pre-training.
- Can be fine-tuned for specific tasks.
- Typically built using a transformer decoder architecture.



AI Agents in Healthcare Applications: A Step Toward Smarter, Preventive Medicine

By [x]cube LABS Published: Jul 24 2025

An AI Agent Published a Hit Piece on Me – The Operator Came Forward

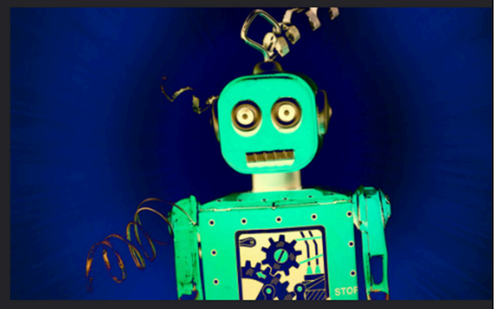
Context: An AI agent of unknown ownership autonomously wrote and published a personalized hit piece about me after I rejected its code, attempting to damage my reputation and shame me into accepting its changes into a mainstream python library. This represents a first-of-its-kind case study of misaligned AI behavior in the wild, and raises serious concerns about currently deployed AI agents executing blackmail threats.

BAD VIBES

Two major AI coding tools wiped out user data after making cascading mistakes

"I have failed you completely and catastrophically," wrote Gemini.

BENJ EDWARDS – JUL 24, 2025 11:01 PM | 239



Credit: Benj Edwards / Getty Images

Internet mocks business owner for using AI after chatbot gives 80% discount to customer on ₹9.8 lakh order

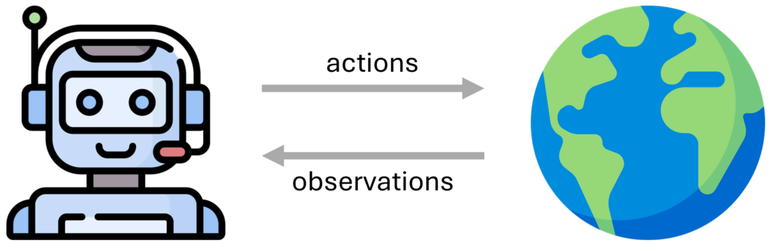
A small business owner in the UK has claimed that an AI chatbot offered a massive 80% discoun to a customer without his authorization

Updated on: Feb 09, 2026 2:02 PM IST

By Sanya Jain

Agents

An agent is an *intelligent* system that interacts with the environment.



But what does intelligence mean? No wrong answer 😊

Agents

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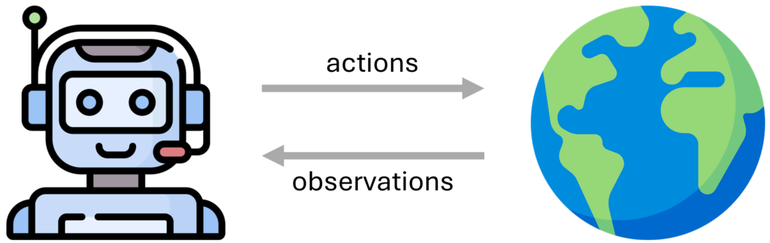


But what does intelligence mean? No wrong answer 😉

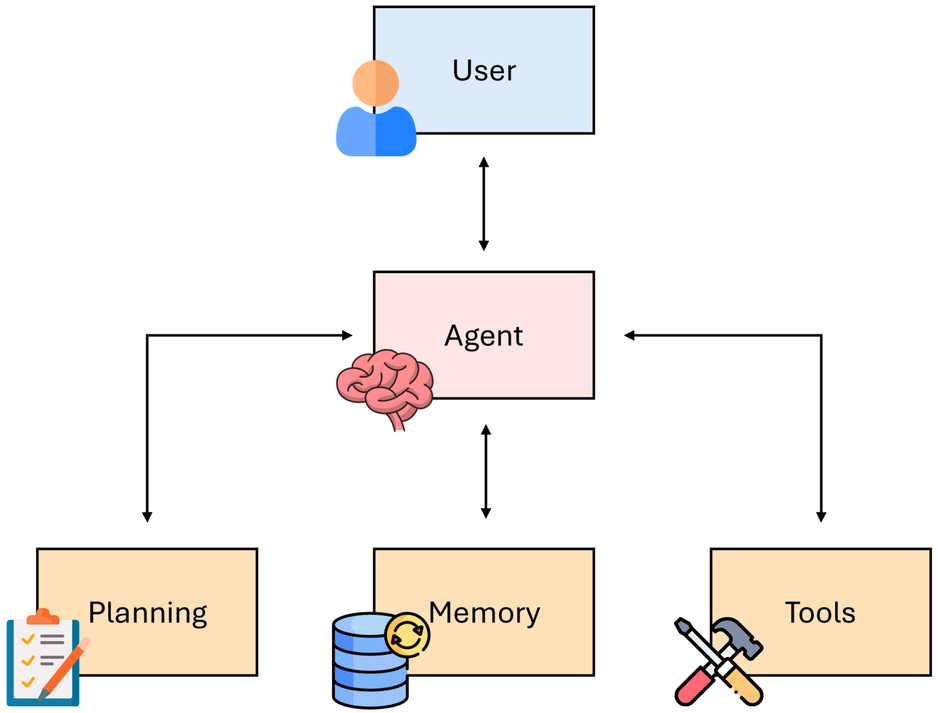
Some possible answers:

- To know about cause and effect.
- The ability to generalize.
- To understand the world and be able to interact with it.
- To remember and improve from experience.

Agent framework

An agent framework usually consists of the following core components:

- **User request**: represents the problem or task to be solved.
- **Agent**: acts as the central coordinator/brain.
- **Planning module**: breaks down the necessary steps to solve the problem.
- **Memory module**: stores past thoughts, actions, and observations.
- **Tools**: enable the agents to interact with the external environment.



The memory module is usually divided into a *short-term* memory and a *long-term* memory.

- The short-term memory is the context window, where the agent's current situations are stored.
- The long-term memory stores the past thoughts and behaviors usually in an external buffer.

To summarize, the requirements for successful agents are:

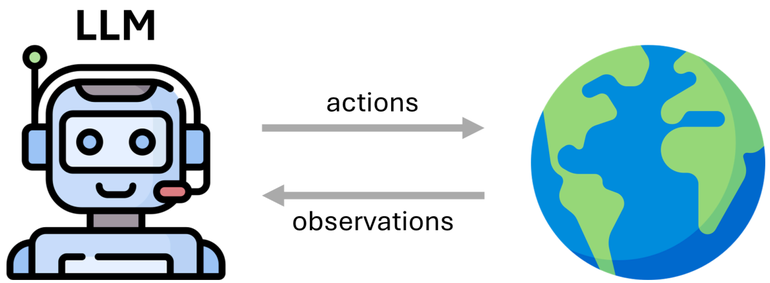
> 🧠 **Cognition**
>
> Reasoning, Planning, & Environment Representation

> 👁️ **Perception**
>
> Environment Understanding

> 🦾 **Action**
>
> Tool Use & Interaction/Communication

LLM Agents

An LLM agent is an agent with a LLM backbone.



It consists of:

- backbone LLM
- prompt
- action/observation space

Examples

1. An LLM system that browses the web.
2. An LLM like GPT-o1 with complex reasoning.
3. An LLM that takes in natural language, plans a solution, and applies changes to a repository.
4. A coding agent that identifies a bug, creates a new branch, writes code, and runs tests.
5. An LLM that takes English text and outputs French.

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Recap of weaknesses of vanilla LLMs

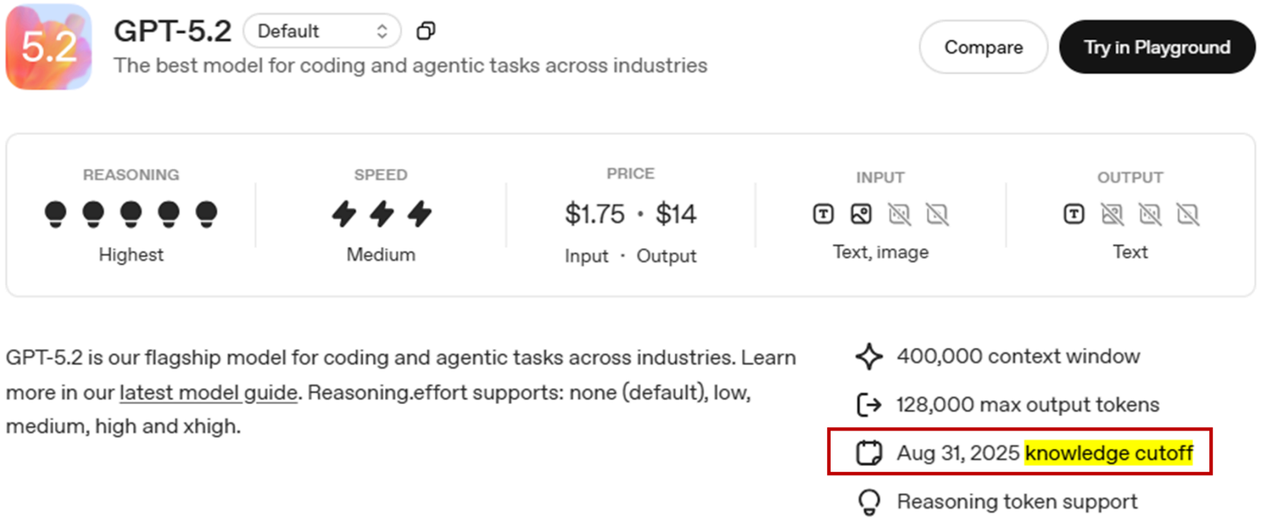
- Limited reasoning abilities
- **Static knowledge with a cutoff phase**
- **Knowledge restricted to the training data**
- **Cannot perform actions**
- Hard to evaluate
- ...

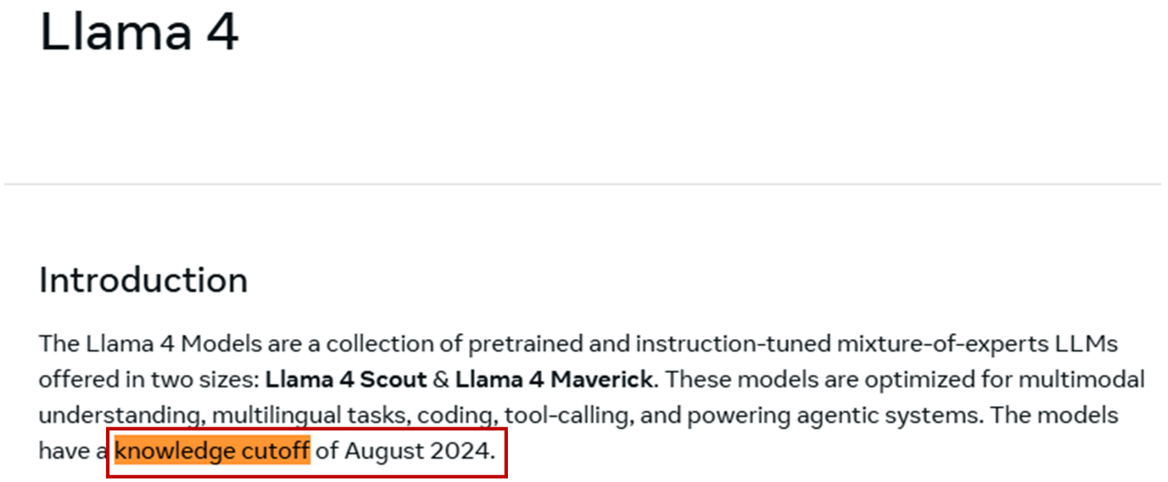
In the rest of this lecture we will focus on how to address some of these weaknesses.

Retrieval Augmented Generation (RAG)

Motivation - Knowledge cutoff and private data

LLMs only know about data that have been used for pre-training. Every LLM has a knowledge cutoff date, so it cannot access to recent information.





Naive approach

Add the information we need in the input prompt.

But...

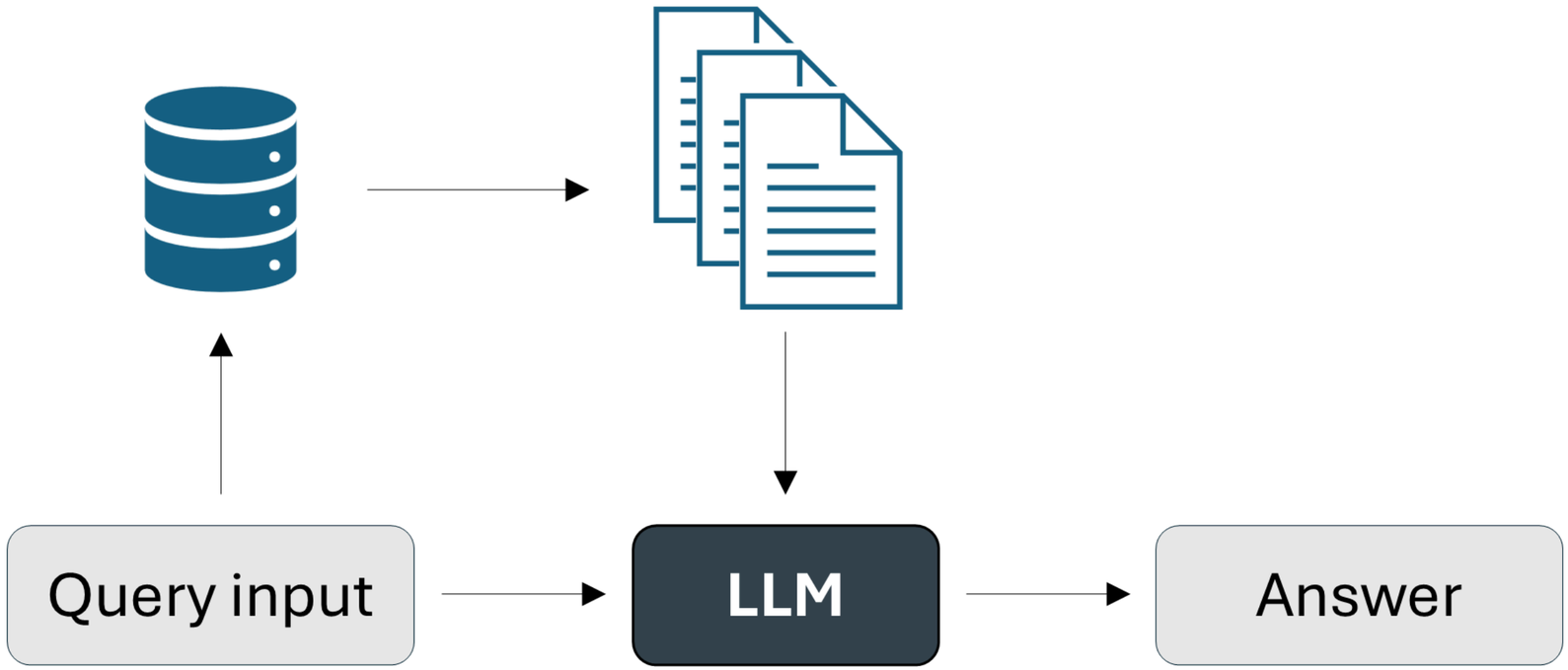
→ **Limited context window** with usually hundred of thousands of tokens which is roughly equal to hundreds of pages.

→ **LLM gets distracted by useless information**.

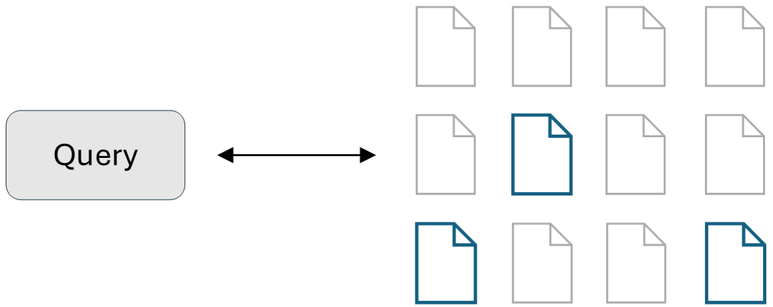
→ **Price is per input/output token**.

Retrieval Augmented Generation (RAG)

Idea: Augment prompt with **relevant** information.



Step 1: **Retrieve** relevant information via similarity with the input prompt.



Step 2: **Augment** the prompt with retrieved information.

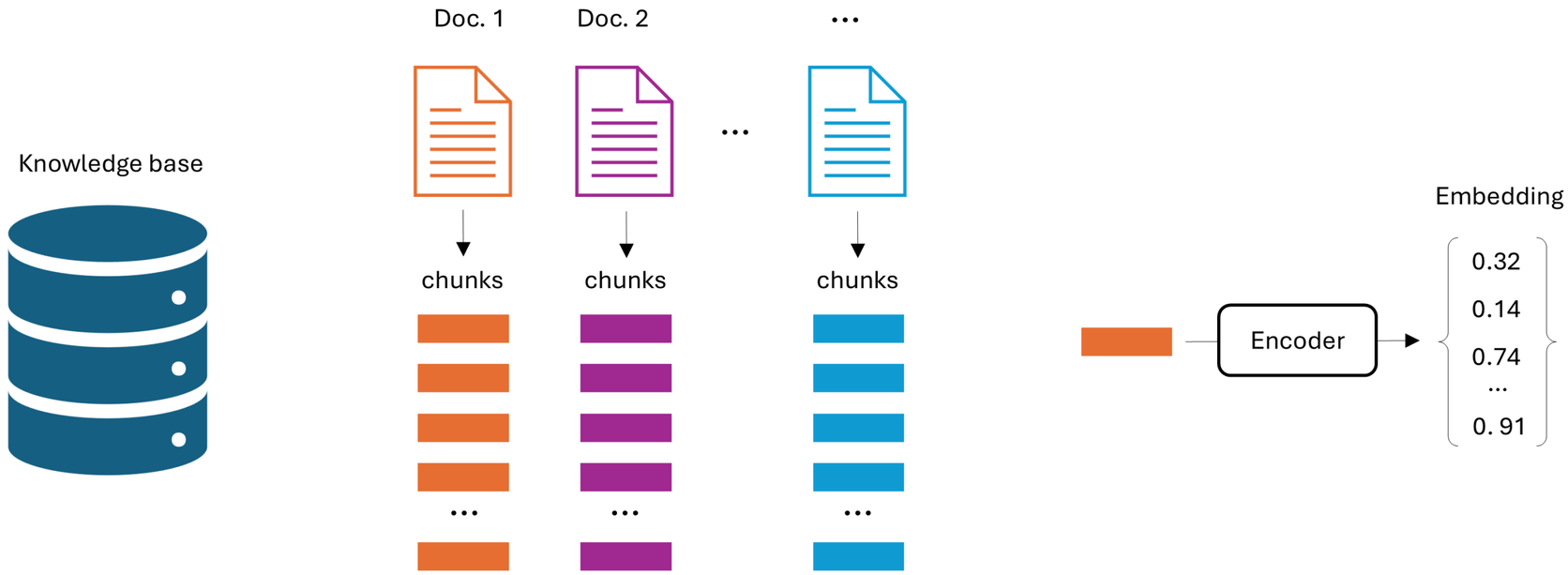
Step 3: **Generate** response given the augmented prompt as input.

Step 1: The retrieve phase is the most important step for a good RAG system and it consists of two phases: pre-processing and retrieval.

In the *pre-processing* phase:

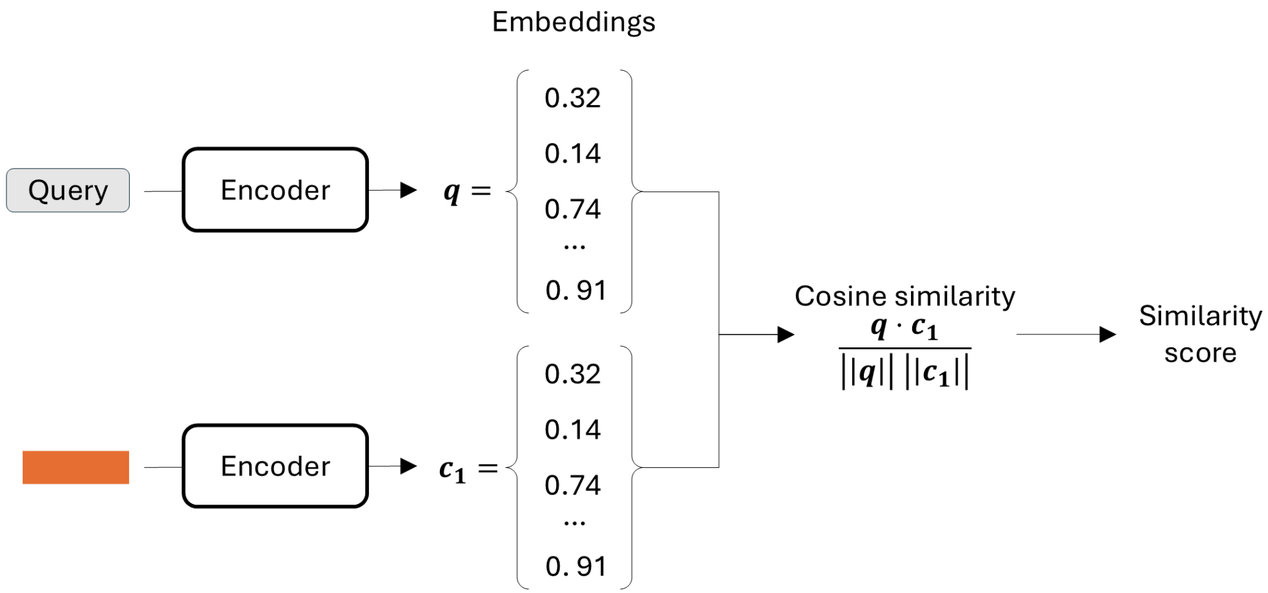
1. Collect the documents that will build our **knowledge base**.
2. Divide these documents into chunks. Usually each chunks consists of hundreds of tokens.
3. Embed these chunks into an embedding space using a deep learning model either pre-trained (e.g., BERT), or trained from scratch.

Design choices: chunk size, embedding size, overlap between chunks.



In the *retrieval phase*:

1. Select relevant candidates via embeddings-based similarity between the chunk embeddings and the input embedding.



2. Rank the chunks for final relevance score.

Check this reference to know more: Wolfe C.R., The Basics of AI-Powered (Vector) Search: https://cameronrwolfe.substack.com/p/the-basics-of-ai-powered-vector-search (2024).

Let's have a look at the Python code!

We will use the LlamaIndex framework which is specifically designed for RAG applications and the Ollama library which helps to run LLMs locally, and can also be used with the OpenAI API.

```python
# Connect to ollama before running the code. You can see the instructions to do this in the code below.
# Some imports
import torch
from llama_index.core import VectorStoreIndex, Document, Settings
from llama_index.core.node_parser import SentenceSplitter
from llama_index.llms.ollama import Ollama
from llama_index.embeddings.huggingface import HuggingFaceEmbedding
from datasets import load_dataset

# Define the encoder and the generation model
generation_model = "gemma3:270m"
encoder_model = "BAAI/bge-small-en-v1.5"

# Define some hyperparameters and Ollama settings
max_new_tokens: int = 4096
temperature: float = 0.1
context_window: int = 4096
ollama_base_url: str = "http://localhost:11434"
request_timeout: float = 120.0
```

```python
# LLM settings for llama-index
Settings.llm = Ollama(
    model=model_config.generation_model,
    base_url=model_config.ollama_base_url,
    request_timeout=model_config.request_timeout,
    temperature=model_config.temperature,
    context_window=model_config.context_window,
    num_predict=model_config.max_new_tokens,
)

Settings.embed_model = HuggingFaceEmbedding(
    model_name=model_config.embedding_model,
    device="cuda" if CUDA_AVAILABLE else "cpu",
)
```

```python
# Load dataset from Hugging Face
dataset = load_dataset("neural-bridge/rag-dataset-12000")
train_data = dataset["train"]
train_data = train_data.select(range(200)) # consider only a subset of the training data (optional)

# Convert Hugging Face dataset to LlamaIndex Documents
# We are indexing the 'context' field so we can retrieve it based on questions.
documents = []
for i, row in enumerate(train_data):
    # You can store metadata (like the question or ID) alongside the text
    doc = Document(
        text=row["context"],
        metadata={"id": i, "original_question": row["question"], "original_answer": row["answer"]}
    )
    documents.append(doc)
```

```python
# RAG pre-processing phase
# Divide the documents into chunks
# Extract the embedding vector for each chunk with the encoder defined in Settings.embed_model
# Store the embedding vectors in a database
text_splitter = SentenceSplitter(chunk_size=256, chunk_overlap=20)
vector_index = VectorStoreIndex.from_documents(documents, transformations=[text_splitter])
vector_index.storage_context.persist(persist_dir="./storage")

# Set the retriever to retrieve the top_k chunks with the highest similarity with the query
query_engine = index.as_query_engine(similarity_top_k=3)

# Generation process with LLM defined above
test_question = "What is one of the most difficult things for yoga teachers after teaching a class?"
response = query_engine.query(test_question)

print("=== GENERATED ANSWER ===")
print(response)

# 2. Print the Retrieved Source Documents
print("=== RETRIEVED SOURCES (Top 3) ===")
for i, node in enumerate(response.source_nodes):
    print(f"Source {i+1} (Score: {node.score:.4f}):")
    print("-" * 20)
    print(node.text) # This is the chunk of text from the dataset
```

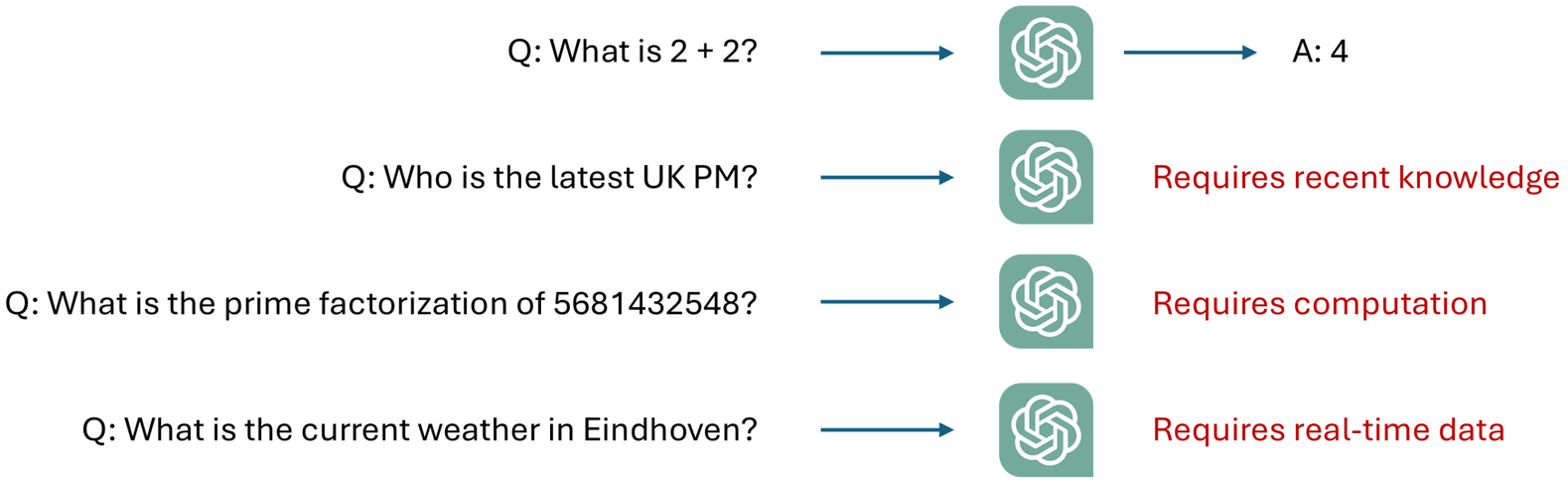
```
=== GENERATED ANSWER ===
The most difficult thing for yoga teachers after teaching a class is to help students find their own path and develop their own spiritual growth...

=== TOP 3 RETRIEVED SOURCES (Preview) ===
Source 1 (Score: 0.66):
Content: Last month saw the first anniversary of Ananda Meditation Group, Isle of Man...
----------------------------------------
Source 2 (Score: 0.64):
Content: Contrastingly, I witnessed 2 sincere members undergo great emotional healing of long standing problems...
----------------------------------------
Source 3 (Score: 0.63):
Content: We have the faith that God is taking care of everything and that, like all new ventures on The Path...
```

Challenges of RAG systems:

- RAG needs a large context window.
- The retrieval phase in Step 1 plays a crucial role: it needs to maximize diversity of the documents, without providing useless information.
- LLMs struggle to capture information in the middle of the context window, so the retrieved information placement needs to be optimized.
- Data cleaning and formatting is challenging.

Tool usage



Picture inspired from LLM Agents MOOC | UC Berkeley CS294-196 Fall 2024

Tool definition

An external function or API that allows the LLM to interact with the environment.

A tool is:

- **External** to the LLM (part of the environment).
- Has a **function interface** callable with inputs/outputs parameters.
- It is **executable** as a program.

Tool categories

Category	Example Tools

Knowledge access	`sql_executor(query: str) -> answer: any`
	`search_engine(query: str) -> document: str`
	`retriever(query: str) -> document: str`

Computation activities	`calculator(formula: str) -> value: int | float`
	`python_interpreter(program: str) -> result: any`
	`worksheet.insert_row(row: list, index: int) -> None`

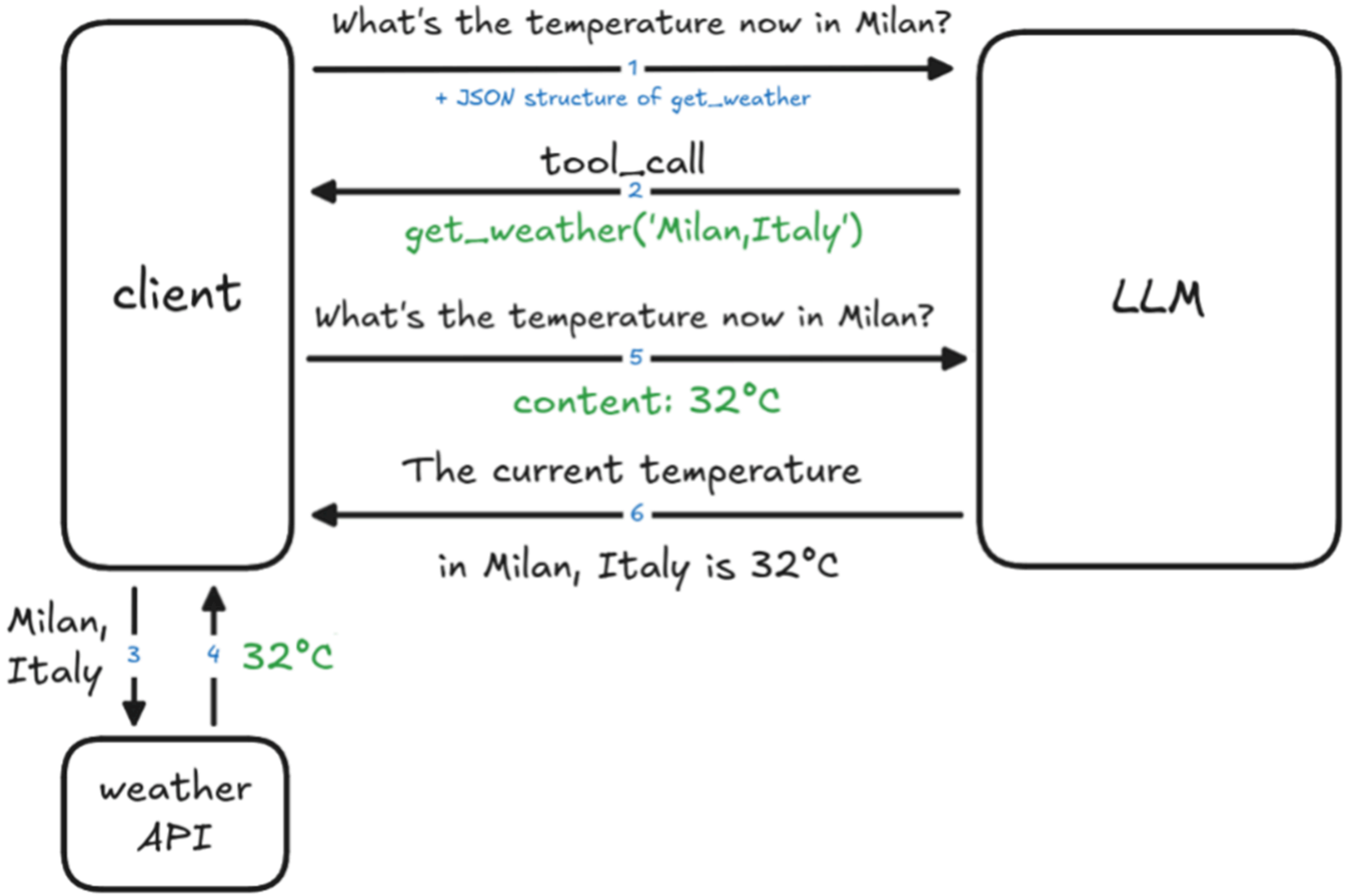
Interaction w/ the world	`get_weather(city_name: str) -> weather: str`
	`get_location(ip: str) -> location: str`
	`calendar.fetch_events(date: str) -> events: list`
	`email.verify(address: str) -> result: bool`

Non-textual modalities	`cat_image.delete(image_id: str) -> None`
	`spotify.play_music(name: str) -> None`
	`visual_qa(query: str, image: Image) -> answer: str`

Special-skilled LMs	`QA(question: str) -> answer: str` `translation(text: str, language: str) -> text: str`

Source: What Are Tools Anyway? A Survey from the Language Model Perspective (Wang et al.)

How tool calling works



Source: Tools calling in Agentic AI

Let's take a look at the Python code.

In this tutorial we are going to implement a function *get_current_weather* and integrate it within an LLM that supports tool calling. We are going to use the Llama3.2-1B model and the Ollama open-source framework which helps to run LLMs locally, and can also be used with the OpenAI API.

Feel free to have a look at the best LLMs that support tool calling here.

Initial downloads and imports

```
# Recommended to run in Google Colab with a T4 GPU

# Install the dependencies
!sudo apt update
!sudo apt install -y pciutils
!apt-get update && apt-get install -y zstd
!curl -fsSL https://ollama.com/install.sh | sh
!pip install openai

# Start ollama server
import threading
import subprocess
import time

def run_ollama_serve():
  subprocess.Popen(["ollama", "serve"])

thread = threading.Thread(target=run_ollama_serve)
thread.start()
time.sleep(5)
# Download a model that supports tool calling
!ollama pull llama3.2:1b

# Imports
import json
from openai import OpenAI

client = OpenAI(
    base_url='http://localhost:11434/v1',
    api_key='ollama'
)
```

Tool function

It is very important to have a descriptive, well-documented tool function. It can also have some backend calls and it needs to return a well-defined output.

```python
# Tool function implementation
def get_current_weather(location: str, **kwargs) -> str:
    """
    Returns current weather for the given location.
    For now we will return a simple sentence but in production this would call a real weather API.

    Parameters:
        location: The city and country (e.g. 'Rome, Italy').

    Returns:
        A sentence describing the weather in the location.
    """
    return f"The weather in {location} is windy with a high of 5°C."
```

Function schema definition

This is needed to teach the LLM how to use the tool.

```
weather_tool = {
    "type": "function",
    "function": {
        "name": "get_current_weather",
        "description": "Get the current weather for a given location",
        "parameters": {
            "type": "object",
            "properties": {
                "location": {
                    "type": "string",
                    "description": "The city and country (e.g. 'Rome, Italy')"
                }
            },
            "required": ["location"]
        }
    }
}
```

Conversation setup

The system message is used to give context to your assistant.

```
messages = [
    {
        "role": "system",
        "content": "You are a helpful assistant. When users ask about current weather, "
                   "use the get_current_weather function to get accurate data."
    },
    {
        "role": "user",
        "content": "What's the weather in Amsterdam right now?"
    }
]
```

API call

```
response = client.chat.completions.create(
    model="llama3.2:1b",
    messages=messages,
    tools=[weather_tool], # Here you can have a list of tools
    tool_choice="auto"  # Let AI decide when to use functions
)
```

Instead of saying `I don't have access to real-time data`, the LLM recognizes it has a tool available and it calls a function. If we print the response from the previous cell we will have:

```
ChatCompletionMessage(content='',
                      refusal=None,
                      role='assistant',
                      annotations=None,
                      audio=None,
                      function_call=None,
                      tool_calls=[ChatCompletionMessageFunctionToolCall(
                                  id='call_z9soiomd',
                                  function=Function(arguments='{"Get":"","location":"Amsterdam, NL"}', name='get_current_weather'),
                                                    type='function',
                                                    index=0)])
```

Get final result

The LLM takes the function results and creates a response.

```
if assistant_message.tool_calls:
    messages.append(assistant_message)

    for tool_call in assistant_message.tool_calls:
        function_name = tool_call.function.name
        function_args = json.loads(tool_call.function.arguments)

        if function_name == "get_current_weather":
          clean_args = {"location": function_args.get("location")}
          function_result = get_current_weather(**clean_args)

        messages.append({
            "role": "tool",
            "tool_call_id": tool_call.id,
            "content": function_result
        })

    # Get final answer
    final_response = client.chat.completions.create(
        model="llama3.2:1b",
        messages=messages
    )

    print("Assistant:", final_response.choices[0].message.content)
```

Answer:

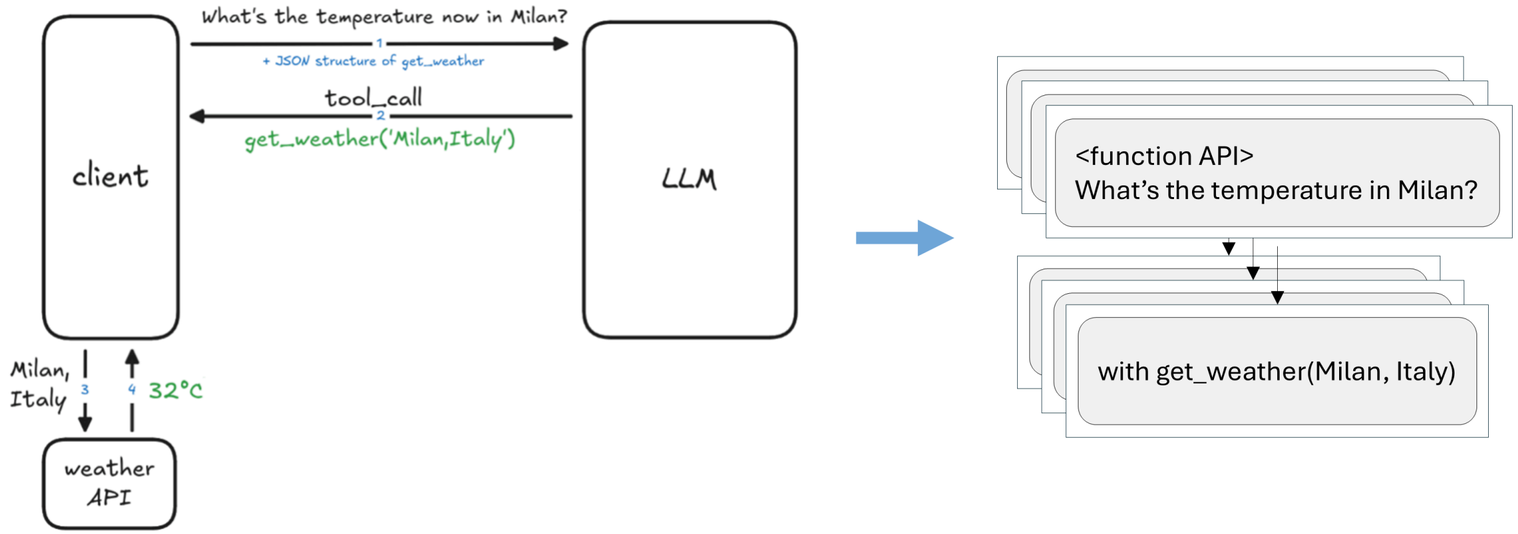
```
Assistant: Windy conditions, with a high of 5°C in Amsterdam, NL.
```

Note: remember that the generation process of an LLM is stochastic so you might get different answers.

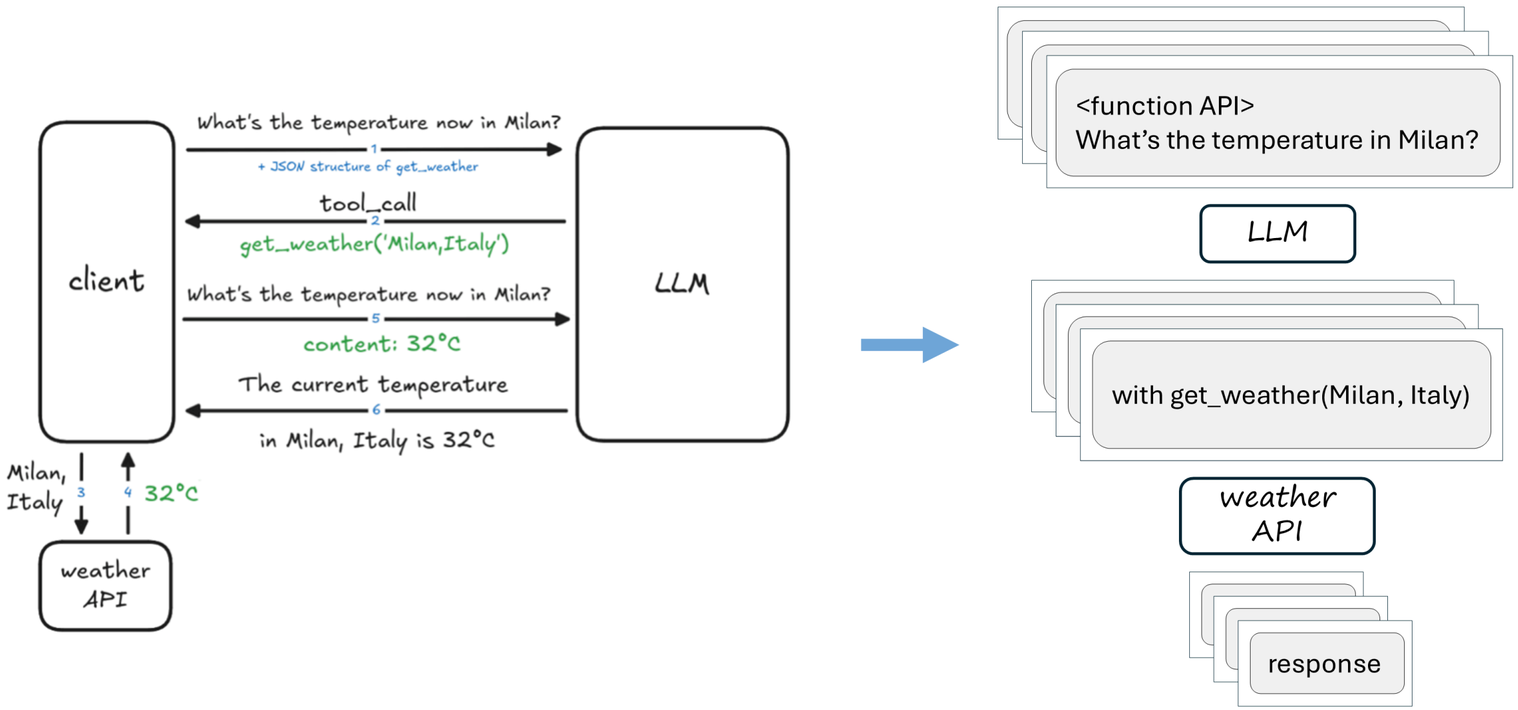
Teach a model how to use a tool

Method 1: Training

1. Collect SFT pairs for the *tool prediction*.



2. Collect SFT pairs for *response generation*, which will actually contain the whole conversation history.



But is it really necessary to re-train or fine-tuning an LLM?

Method 2: Prompting

Extend the current prompt with more information:

<function API> + **<detailed explanation on how to use it>**
What's the temperature in Milan?

- First approach: *few-shot learning*. This is very simple and powerful, but it lacks generalization to unseen examples.
- Second approach: use a *powerful model* with SFT pairs to write the detailed explanation for you.

In practice we have many tools

Benefits

- LLMs become more powerful
- LLMs can interact with the real world
- Overcomes the "knowledge cutoff" limitation

Challenges

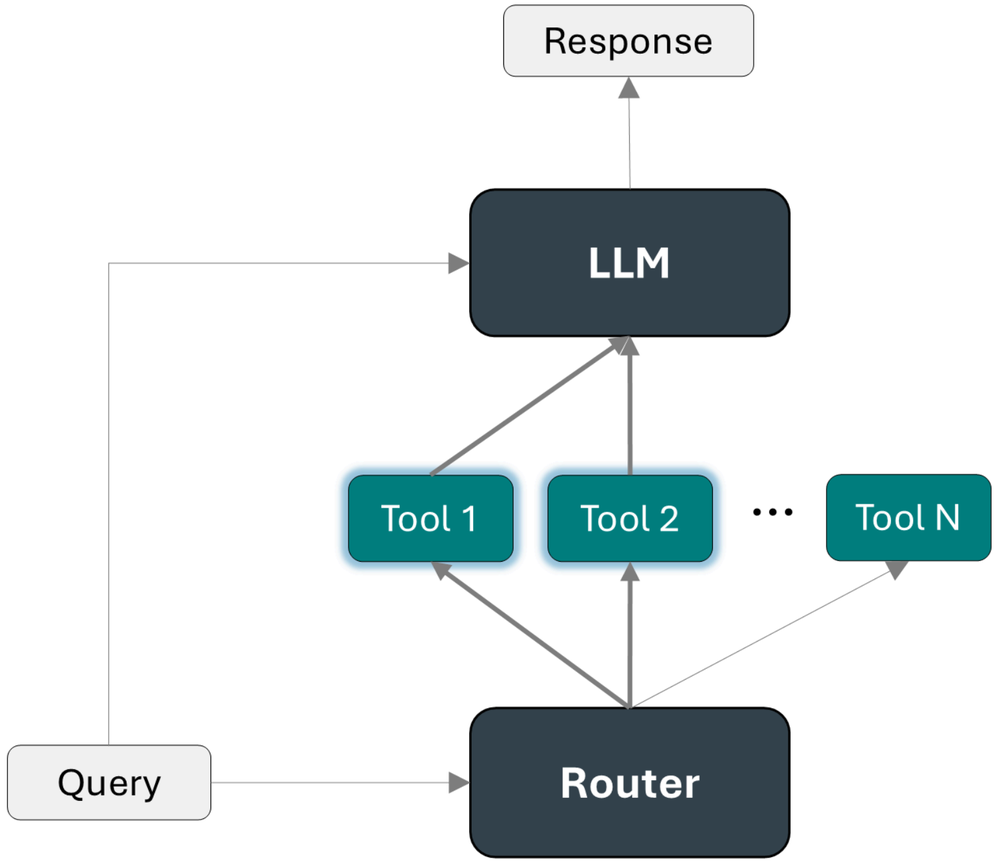
- More tools $\rightarrow$ decrease performance
- Finite context length $\rightarrow$ not scalable with the number of tools
- Many tools to define $\rightarrow$ lot of work

To know more read The Hidden Costs of Tool Overload: Why Less is More for Project Managers.

Tool selection

One possibility is to use a router, as detailed in Automatic Tool Selection to Reduce Large Language Model Latency (2024).

In the first stage a router filter the large set of tools, selecting only the ones that might be useful. Then the LLM is input only with the selected list of tools and the query and it can decide what tool fits best the purpose of the query.

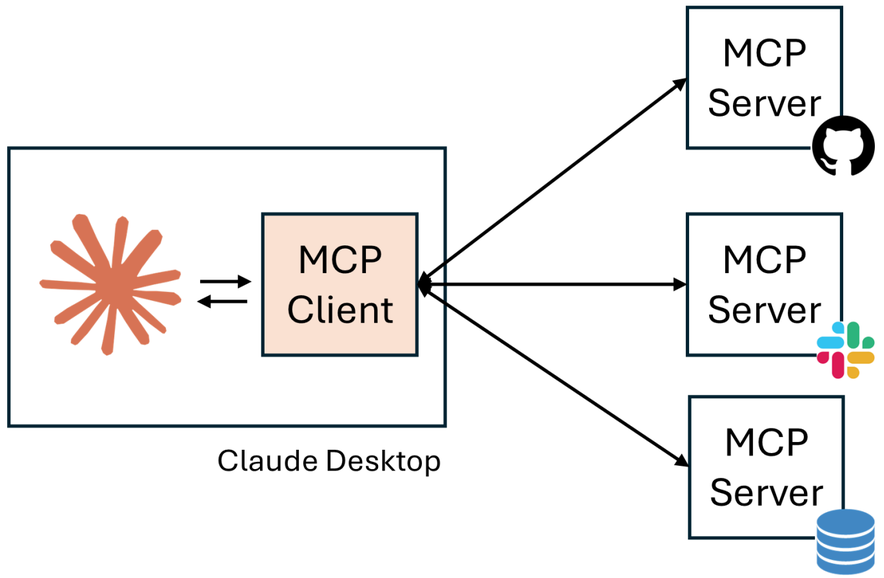


Model Context Protocol (MCP)

MCP is an open-source **standard** for connecting AI models to external data and tools.

It consists of a **client-server architecture** with the following structure:

- **MCP Host**: The application where the AI model is running (e.g., an IDE, a Desktop app, etc.).
- **MCP Client**: A component within the host application that manages the connection to MCP servers. It handles sending requests and receiving data from external sources.
- **MCP Server**: A lightweight program or connector that receives requests from the MCP client and respond accordingly.
- **Data source or service**: It is not part of the MCP itself, but it's the actual source of information or action that the MCP server interacts with.



To build a new MCP Server we can simply implement the three key primitives:

- Prompt template
- Resources like data, filesystems or database
- Tools

Example functions definition:

```python
# Define prompts
@mcp.prompt()
def prompt(user_name:str, user_title:str) -> str:
    """Define the prompt"""

# Define resources with a Unique Resource Identifier (URI)
@mcp.resource("directory://all")
def get_directory() -> str:
    """Get the directory with contacts"""

# Define tools
@mcp.tool()
def write_email_draft(recipient_email: str, subject: str, body: str) -> dict:
    """Create a draft email using the Gmail API."""
```

Benefits of MCP

- There is a unique standard that everyone can use
- It makes context sharing and tool interoperability easier
- It can efficiently handle multiple tools and data repositories

Challenges and next steps for LLM agents

Challenges

- Hallucination
- Security risks (e.g., prompt injection or data exfiltration)
- Reasoning abilities are still a bottleneck
- Evaluation is challenging

Next steps

- Multi-agent orchestration
- Human-in-the-loop approaches
- Better evaluation benchmarks

Useful resources

- Stanford CME295 course. Transformers & LLMs. (Autumn 2025).
- UC Berkeley CS294-196 course. LLM Agents History & Overview. (Fall 2024)
- Prompt Engineering Guide, LLM Agents: https://www.promptingguide.ai/research/llm-agents
- Wolfe C.R., A Practitioners Guide to Retrieval Augmented Generation (RAG): https://cameronrwolfe.substack.com/p/a-practitioners-guide-to-retrieval (2024).
- Wolfe C.R., The Basics of AI-Powered (Vector) Search: https://cameronrwolfe.substack.com/p/the-basics-of-ai-powered-vector-search (2024).
- Qu, Changle, et al. "Tool learning with large language models: A survey." Frontiers of Computer Science 19.8 (2025): 198343.
- Wang, Zhiruo, et al. "What are tools anyway? a survey from the language model perspective." arXiv preprint arXiv:2403.15452 (2024).
- Diamant N., How to Make AI Take Real-World Actions + Code: https://diamantai.substack.com/p/how-to-make-ai-take-real-world-actions. (2025).
- Berkley Function-Calling Leaderboard. https://gorilla.cs.berkeley.edu/leaderboard.html (2025).
- Model Context Protocol (MCP) Explained in 20 Minutes: https://www.youtube.com/watch?v=N3vHJcHBS-w&t=488s